

SALES PROGRESSOR

Feature and functionality reference

A complete capabilities inventory of what agents, sellers, and buyers get from the app.

Date 3 June 2026

Reflects commit `3dea586`

Source audited from the codebase, not from product copy

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How to read this document

A capabilities inventory built by reading the salesprogressor codebase, not by paraphrasing the marketing site.

The vocabulary is the product's own. A piece of work in progress is a **sale** or a **file**, never a "transaction" (the code uses that word but the customers do not). The discrete pieces of progress on a sale are **steps**, never "milestones" (the code uses that word too). The product is **Sales Progressor** or **the app**, never a "platform".

Prices come from `lib/billing/fee.ts`. Sales Progressor is not VAT registered today, so every fee quoted is the actual headline price. Charging fires on **exchange**, not on completion.

Two callouts are used through the document:

- **SHIPS NOW** The capability is live in the codebase at commit `3dea586`.
- **PLANNED** The schema or scaffolding exists but the capability is not yet wired through to users.
- **INTERNAL** The capability is for Sales Progressor's own staff, not agents or clients.

The end of the document lists items the audit surfaced but could not fully confirm from the code. Treat that appendix as the to-do list for the next pass.

Part 1 - What each audience gets

Three audiences run through the product. Agents work in the full app; sellers and buyers experience it through the portal and the emails.

1.1 Agents

Agents are customer-agency staff: **directors** and **negotiators** at an estate agency that bought into the app. They sign in at `portal.thesalesprogressor.co.uk` and land in the agent app at `/agent/*`. The agent app is the full product surface.

Sidebar nav

Every agent gets the same primary menu (some items are role gated):

Menu item	Route	What it does
Hub	<code>/agent/hub</code>	The home page. Pipeline health, what needs attention today, momentum.
Reminders	<code>/agent/work-queue</code>	The day's chase list. Hidden for the internal admin role.
Completions	<code>/agent/completions</code>	Exchanged files tracking through to completion day.
To-Do	<code>/agent/to-do</code>	Manual tasks the agent or the progressor team have queued.
Updates	<code>/agent/comms</code>	A feed of every step confirmation across the pipeline.
Auto emails	<code>/agent/automated-emails</code>	Pending, sent, errored, and forecast client chase emails.
All Files / My Files	<code>/agent/transactions</code>	The full list of sales, with status tabs and time-based filters.
Analytics	<code>/agent/analytics</code>	KPI dashboard and pipeline charts.
Partners	<code>/agent/partners</code>	Solicitor directory and broker settings.
Account	<code>/agent/account/profile</code>	Name, email, notification preferences.
Billing	<code>/agent/account/billing</code>	Director only. Payment method, terms acceptance, invoice history.

The Hub SHIPS NOW

The agent's home page assembles a daily snapshot of the business. The strip across the top shows four pipeline-health figures:

- **Active files** - count of sales in the active state.
- **Exchanging soon** - sales whose expected exchange date falls in the next 30 days.
- **Need attention** - the count of unique files with reminders that are due today, overdue, or escalated.
The figure turns red when any reminder is escalated, amber otherwise.

- **Pipeline value** - the sum of purchase prices across all active sales.

Beneath that strip, the hub renders:

- A **Needs your attention** card listing up to three of the day's most urgent reminders.
- A **Coming up** strip: how many sales are exchanging this week, completing this week, and the value closing this month.
- A **Stalled files** warning: how many files have not seen a step completion in 14 or more days.
- A **Momentum ring** showing pace versus the previous month.
- An **Exchange forecast** showing exchange volume across the next four weeks.
- A **Service split** donut: how many of the active files are self managed versus outsourced to the Sales Progressor team.
- A **Recent activity** feed of the last few step confirmations.
- A **Diary** showing today's expected exchanges and completions.
- An **Unassigned files** widget for directors (files without an owner).
- An **Expired holds** widget that surfaces on-hold files past their planned end date.

An empty-state variant appears when the agent has no active sales.

All Files / My Files SHIPS NOW

The same route renders one of two titles depending on visibility. Directors and admin see **All Files** across the agency; negotiators see **My Files** scoped to their own sales unless their director has flipped the `canViewAllFiles` flag. The list supports:

- Status tabs: active, on hold, completed, withdrawn, draft.
- Time-based filters: exchanging this week, completing this week, closing this month, exchanging in the next 30 days (the hub strip cards deep-link into these filters).
- Free-text search across property address.
- Columns: address, progress percentage, expected exchange date, owner, status.
- An **Exchange forecast** month-pill strip across the top.
- A **New sale** call to action.

The sale detail page SHIPS NOW

Every individual sale gets its own page at `/agent/transactions/[id]` with a header strip, a sidebar, and a tabbed body.

The **header** shows the property address, the agency, the sale status (active, on hold, completed, withdrawn), the tenure (freehold or leasehold), the purchase type (mortgage, cash buyer, cash from proceeds), and a service-type pill (Self managed or With progressor). Admins can click the service-type pill to flip a file mid-flight; the pill becomes interactive only for them. A back link returns to the file list.

The **sidebar** holds:

- Purchase price, tenure, purchase type.
- Predicted completion date.
- Agent fee, referral fee, broker referral fee.
- Key dates strip.

- A **file time tracker** showing how many hours the agent and the progressor team have spent on this file.
- The **next step** widget.
- Exchange and completion status badges.

The body is tabbed:

- **Overview** - contacts (vendor, purchaser, solicitors, broker), next step, active reminders, recent activity, the property chain, risk score, property intelligence, and internal notes.
- **Steps** - vendor and purchaser step panels side by side. Each row shows the step's friendly name, status (complete, available, locked, not required), the date it was completed, grace days remaining if a reminder is attached, and a client-chase state chip if the system is chasing a client on this step.
- **Reminders** - the list of reminder rules attached to this file, their due dates, status, snoozed state, chase task counts, and controls for marking done, snoozing, waking, or chasing. A pause toggle stops automated client emails for this file without affecting the agent reminder.
- **To-Do** - manual tasks specific to this file, including "ask the progressor" requests from agents to the internal team.
- **Activity** - a timeline of step completions, internal notes, contact edits, and automated email logs.

Reminders (the work queue) SHIPS NOW

The work queue is what an agent opens first thing in the morning. It groups the day's reminders by urgency:

- **Escalated** in red - the system has chased the limit and not heard back.
- **Overdue** in orange.
- **Due today** in amber.
- **Coming up** across the next few business days.

Each row shows the property address, the reminder name, the next due date, the count of times this has been chased, and a fallback-kind chip if the automated client chase could not run (for example the contact has no email on file, or the file has automated chases paused). From the row the agent can mark done, snooze, wake, or record a chase. A header strip shows stat pills (X overdue, Y due today, Z coming up).

Completions SHIPS NOW

After exchange a file leaves the day-to-day work queue and lives in Completions. The page groups exchanged files by completion urgency: overdue, this week, next week, later, no date set. Each row carries the address, completion date, purchase price, the fees billed on this file, and both solicitor firms. A summary line totals the file count, fees, and total sale value.

Updates SHIPS NOW

An app-wide feed of every step that has been confirmed. Grouped by day (Today, Yesterday, This week, Last week, and so on), then by file. Each row shows the date and time, the step name, the side (Sale or Purchase), who confirmed it, and a portal-confirmation badge when the client did the confirming. A filter toggle switches between "All steps" and "Client confirmations only".

Auto emails SHIPS NOW

A full feed of the automated client chase system. The page is scoped per role and shows four buckets:

- **Pending** - emails queued to send.
- **Sent** - the last 30 days of delivered chases.
- **Errored** - sends that failed (bounced, no email on file, etc.).
- **Upcoming forecast** - the next 14 days of expected chases.

For each row the agent sees the file, the email status, the recipient, the send time, and the error reason if any. Directors can scope to "mine only" or the whole agency.

To-Do SHIPS NOW

Manual tasks separated into:

- **My to-dos** - tasks the agent has created for themselves.
- **With progressor / From agents** - the cross-team request inbox. An agent on an outsourced file can flag a question to the progressor team and the progressor sees it here.
- **Overdue** - the running list of things past their due date.

Partners SHIPS NOW

Two halves to one page:

- **Solicitor directory** - every solicitor firm contacted on an active file, grouped by firm, with contact details, active file count, referral count, and chips to jump straight to the file.
- **Broker settings** (director only) - preferred mortgage broker firm, contact, website, referral fee; plus a recommended-solicitors list with default referral fee fields.

Analytics SHIPS NOW

KPI cards (files created, exchanged, completed, completion rate, average time to exchange, average fees), an optional team-member filter for directors, and charts for monthly activity, KPI trends, files at risk (stalled 14 or more days), and referral income broken down by solicitor and broker.

New sale SHIPS NOW

A guided form at </agent/transactions/new-v2>. The agent enters:

- Property address.
- Purchase price.
- Tenure (freehold or leasehold).
- Purchase type (mortgage, cash buyer, cash from proceeds).
- Share of freehold flag.
- Vendor and purchaser solicitor selections.
- Mortgage broker (if applicable).
- Service type (self managed or outsourced to Sales Progressor).
- Notes.

If outsourced is picked the form adds an assignment step. The director-level billing gate blocks the form if the agency's trial has ended and no payment method is on file. Internal staff bypass this gate.

Account and notifications SHIPS NOW

At `/agent/account/profile` an agent can change their name, email, and email-notification preferences (weekly brief, chain notifications, and similar opt-outs).

Director-only surfaces SHIPS NOW

Directors get extra controls that negotiators do not see:

- **Billing** at `/agent/account/billing`. Payment-method capture via Stripe, terms acceptance, the current month's running total, this month's trial savings, pending credits, and a history of issued invoices.
- **Automation settings** at `/agent/settings/automation`. The master toggle that switches the automated client chase on or off for the agency, plus per-step grace days and repeat-every-days. Hard-coded ceilings of two chases maximum and 14 days of silence cap the engine.
- **Team management** at `/agent/account/team`. Invite negotiators, manage their `canViewAllFiles` flag, resend or cancel pending invites.
- **Mortgage broker and recommended solicitors** settings on the Partners page.

1.2 Sellers

Sellers (vendors in the code) do not get a full Sales Progressor account. They access the app two ways: through their **portal** link, and through the **emails** the app sends them.

What the seller portal gives them SHIPS NOW

The portal lives at `/portal/[token]`. The token is the seller's contact-level token (no password). It is mobile first, sized for a phone, and installable as a progressive web app.

Tab	What the seller sees and can do
Home	Overall progress ring; the next action card (the highest-priority step they can confirm); a "key dates" panel; a "recent activity" snapshot; contextual pro tips; a feedback widget; and a banner once the file has exchanged or completed.
Progress	The full step-by-step checklist grouped by phase. Each row shows status (done, pending, not applicable), the friendly label, who is responsible (you, your solicitor, the agent), the typical duration, the date completed if done, and any event date that has been recorded. Both sides of the chain are visible labelled "Sale" / "Purchase" so each side sees the whole picture.
Updates	A timeline of activity grouped by date: step confirmations and outbound messages from the agent (each carrying a method badge: email, phone, SMS, voicemail, WhatsApp, post).
Respond	The deep-link destination from a chase email. Shows the steps that are currently being waited on. For each row the seller can tap "Yes this is done", "Tell us when" (set an expected date), or "Leave a note" (free text to the agent).
Exchange	Appears once the file has exchanged. A vendor-specific pre-exchange checklist (post redirect, packing, cancelling direct debits), an exchange-day checklist (utility meter reads, keys/fobs/remotes, appliance manuals, service records, clearing the property), and post-exchange guidance (the solicitor's statement, mortgage redemption, capital-gains-tax pointers).
Complete	Appears once the file has completed. Vendor-specific final checklist and what-happens-next guidance.

Actions a seller can take from the portal

- **Confirm a step.** The portal shows a confirm button for steps that are available to the seller. The portal cannot confirm the six agent-only steps (VM18, PM25, VM19, PM26, VM20, PM27).
- **Set an expected date.** When the agent does not yet know when a step will happen, the seller can supply one.
- **Leave a note.** Free text addressed to the agent. Lands in the agent's notifications bell and on the file's activity feed.
- **Upload a survey report.** Up to three files at a time, up to 10MB each, PDF/JPG/PNG/DOCX. Stored to the file in Supabase storage.
- **Explain an email.** The seller can paste a confusing email from their solicitor and the app returns a four-section plain-English breakdown (what they are saying, what you need to do, deadlines, things to flag). Rate limited to three explanations per contact per hour.
- **Install the portal as a PWA.** Web push opt-in is supported.
- **Unsubscribe.** Every chase email carries an unsubscribe link. Clicking it stops chase emails and weekly updates but does not block the portal itself.

What a seller receives by email SHIPS NOW

Email	When	What it does
Portal invite	Agent presses "Send invite" against the contact row	Personal greeting, "track your sale online", a single button to the portal home.
Step confirmation	When the other side of the file confirms a step	"Your solicitor has issued the draft contract pack", and so on. Plain-English description plus context.
Client chase digest	Daily 08:30 cron when one or more steps are sitting on the seller	Subject is address first. Lists the items waiting on the seller. CTA is a single link to the Respond page in the portal.
Client weekly update	Saturdays 09:00	A status email if no recent activity has happened (gentle reassurance).
Exchange celebration	VM19 confirmed (first time per file)	"Contracts exchanged" with the next-steps overview.
Completion survey	Fire-and-forget after VM20 is confirmed	Short feedback survey.
Withdrawal cascade	If the file or a chain partner withdraws	The agent sees this; the seller is told via the agent.

Full descriptions of these emails appear in section 2.1.

What sellers do NOT have

- No login. Token in URL only.
- No editing of their own name, phone, or solicitor contact (these are agent-side fields).
- No direct messaging into the portal beyond "leave a note" on a chase response.
- No download UI for documents they have uploaded (those go through the agent).

1.3 Buyers

Buyers (purchasers in the code) use the same portal surface as sellers, with role-specific copy and a different set of steps. The differences are not cosmetic: the buyer's process is genuinely different.

How the buyer's portal differs SHIPS NOW

- **Language:** "Your purchase of..." instead of "Your sale of...". Tips and checklists rewritten throughout.
- **Steps:** the buyer follows PM1 through PM27 (27 steps), versus the seller's 20.
- **Mortgage-related steps:** PM5 (mortgage application), PM6 (lender valuation), PM11 (mortgage offer) appear only for mortgage buyers. They auto-skip for cash buyers.
- **Searches:** PM8 (orders searches) and PM13 (receives search results) are purchaser-only.
- **Survey:** PM9 (book a Level 2 or 3 survey) is the only step in the whole system that the user can opt to mark as not required from the portal.
- **Deposit:** PM24 (transfer the deposit) auto-skips when the buyer's deposit comes from a concurrent sale's equity (cash from proceeds).

- **Exchange and completion checklists** are rewritten for the buyer: collect keys, take meter readings, buildings insurance, check items left, transferring the balance to the solicitor, broadband, change of address list, post redirect.
- **Post-completion guidance** differs: the buyer is told about Land Registry, Stamp Duty, council tax setup, and confirming insurance is active.

What buyers receive by email

The same set as sellers, with buyer-specific copy. The portal invite, step confirmations, chase digests, weekly updates, exchange celebration, and completion survey are all sent to both sides where applicable. The chase digest tone branches on whether the buyer is being asked to do something themselves ("Yes this is done") or whether the system is nudging on something sitting with their solicitor or lender ("Yes this is already done").

Part 2 - Cross-cutting capabilities

Capabilities that span audiences: emails, reminders, the step engine, the portal, chain handling, billing, accounts.

2.1 Emails sent

Every outgoing email passes through SendGrid. Most lifecycle emails are written to an

`OutboundEmailQueue` table first and a draining cron picks them up during business hours; a few high-priority emails send synchronously. The complete list of outgoing emails follows. Internal-staff-only emails are tagged **INTERNAL** and full descriptions live in Appendix A.

Client and agent emails (the lifecycle set)

Email	Trigger	Recipient	Purpose	Key content
Portal invite	Agent clicks "Send invite" against a contact on the file page	Vendor or purchaser	Get the client into the portal	Personal greeting, "track your {sale/purchase} online", button to portal home, agency name and contact details.
Step confirmation (single)	A step is confirmed on a file	Other contacts on the file (vendor, purchaser, solicitor, broker)	Let everyone on the file know progress has happened	Step name, plain-English description, next step context. One email per step.
Step confirmation (digest)	Two or more confirmations have stacked up for the same recipient inside the 3-minute drain window	Other contacts on the file	Avoid email spam when several steps tick in quick succession	Batched digest of all unsent confirmations.
Client chase digest	Cron at 08:30 UK Mon-Sat. Fires when steps are due and the contact has not engaged with the portal in 14 days	Vendor or purchaser	Move things along when the client is who's holding the file up	Subject is address first. Sectioned by tone: items the client needs to do (DIY) versus items the solicitor or lender is holding (NUDGE). Single CTA to the Respond page. Unsubscribe link.
Client weekly update	Cron at 09:00 UK Saturday	Vendor or purchaser	Reassure the client when nothing has moved recently	Short status summary. No CTA pressure.
Exchange celebration	VM19 or PM26 confirmed (once per file)	Vendor or purchaser on this file, plus the agents on chain-mate files	Tell everyone the contract has exchanged	"Contracts exchanged on {address}". For clients: next-steps overview. For chain mates: address of the file that exchanged.
Completion celebration	VM20 or PM27 confirmed	Same audience as above	Tell everyone the sale has completed	"Completed on {address}".
Whole-chain celebration	Every claimed link in the chain has hit completion	The agents on every claimed file in the chain (one each)	Mark the chain finishing	One-time per chain. Guarded against duplicates.
Completion survey	VM20 confirmed, fire-and-forget	Vendor and purchaser	Get feedback	Short feedback request.
Manual chase email	Agent sends from the file page	Any contact on the file	Agent-composed chase	Free-form, agent-authored.

Explain an email confirmation	Returned to the same browser session	Vendor or purchaser	Plain-English breakdown of solicitor jargon	Not an email send; rendered into the portal in a modal. Listed for completeness.
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Chain notifications

Email	Trigger	Recipient	Purpose	Key content
Chain invite	Agent presses "invite" against a stub link in a chain	The agent at the other end (the counter-party agency)	Pull the other agent into the chain	Originator's agency, property addresses on both sides, chain position, a Claim button and a Decline button. Token-signed URLs.
Chain invite decline notification	Counter-party clicks Decline	The originating agent	Tell the originator the other side has declined	Identity of the recipient who declined.
Chain exchange notification	VM19 or PM26 fires on any chain-linked file	Every claimed chain mate (agent)	Tell the chain that one file has exchanged	Address of the exchanged file.
Chain completion notification	VM20 or PM27 fires on a chain-linked file	Every claimed chain mate	Tell the chain that one file has completed	Address of the completed file.
Lost buyer	A file in the chain transitions to withdrawn	The nearest upward claimed chain mate	Tell the agent above that their buyer has pulled out	Address. Asks for a decision: remarket, wait, or withdraw.
Lost purchase	A file in the chain transitions to withdrawn	The nearest downward claimed chain mate	Tell the agent below that the property has fallen through	Address. Asks for a decision.
Asked to wait	The chain is being reformed after a fall-through	The next claimed mate	Confirm the client is willing to wait	Address. Asks for response.
Wait nudge	14 days after an Asked-to-wait with no response	The same recipient	One-shot reminder	"Still waiting on {address}?" Sent at most once per row.

Account and access emails

Email	Trigger	Recipient	Purpose	Key content
Activation (day 1)	Sign-up completes via password or OAuth	The new user	Welcome and orientation	"Welcome to Sales Progressor", what to do next.
Claim welcome	A new user finishes signing up via a chain claim link	The new user	Welcome customised to the file they just claimed	Property-aware welcome. Deduplicated against the generic activation email via <code>welcomeEmailSentAt</code> .
Stuck on day 3	Retention sweep cron at 09:00 UK	The user	Re-engage users who haven't confirmed a step	Activation nudge.
Quiet at 30 days	Retention sweep cron	The user	Reach out to dormant accounts	Check-in.
First exchange	The agency's first ever exchange	The user who created the file	Celebrate the agency's first exchange	Congratulations message.
Send-to-us drop (21 days)	Retention sweep cron, outsourced files only	The user	Drag a stalled outsourced sign-up over the line	Specific to the outsourced funnel.
Last touch (60 days)	Retention sweep cron	The user	Final retention email before pause	Stop sending after this.
Director invitation	A negotiator invites a director into a new agency	Director recruit	Get a director onto an empty agency	Personal greeting, accept link, 7-day expiry.
Negotiator invitation	A director invites a negotiator	Negotiator recruit	Add an agent to the agency	Personal greeting, accept link, 7-day expiry.
Password reset	User requests a reset	The user	Reset password	One-hour-expiry token URL.
DNS setup instructions	Agent triggers from the email-domain page	Whoever the agent nominates as their DNS admin	Get SPF/DKIM records onto the agency's sending domain	Step-by-step DNS guide.
Domain broken alert	Daily 02:00 check finds an authentication failure	The agency owner	Tell them sending will fall back	Which record broke.

Agent operations emails

Email	Trigger	Recipient	Purpose	Key content
Agent weekly brief	Cron at 07:00 UK Monday	Director or negotiator (per-user opt-in)	Catch the agent up on what needs attention this week	Files needing attention; exchanges and completions approaching; a "you have N files all progressing normally" reassurance line.

Internal staff emails

Daily brief, weekly review, morning digest, medians ready, content batch: see **Appendix A**.

2.2 Reminders and notifications

The reminder engine is the system that turns "files are happening" into "the agent has a single list of what to do today". It is a three-layer cascade:

1. **ReminderRule** - configuration. There is one rule per chase scenario the founder has decided is worth chasing. Each rule has a grace-days value (how long after the anchor date before the first chase), a repeat-every-days value (how long between chases), an escalate-after-chases count, and a flag for whether it requires exchange-readiness.
2. **ReminderLog** - per-file state. The engine creates one log row per applicable rule per active file. The log holds the next due date and the active/snoozed/completed status.
3. **ChaseTask** - the agent's work item. When a log goes due, a task is spawned. The task is what the agent actually interacts with on the work queue.

How a reminder advances

- The engine evaluates every active sale against every active rule each weekday morning at 06:00 UK.
- For each rule, it resolves the anchor date (the completion date of the target step, or the file's creation date as a fallback).
- First due date = anchor + grace days, normalised to 06:00 UK on that calendar day.
- If the file is not yet exchange-ready and the rule requires it, the log stays dormant.
- If the target step is already complete or not required, the log is deactivated.
- When the agent records a chase, the log's next due date advances by the repeat-every-days value.
- Once chase count crosses the escalate-after-chases threshold and the grace window has elapsed since the last chase, the row is flagged Escalated. The agent sees a red row.
- The agent can snooze the row, mark it done, or wake it back up.

Where reminders surface

- **Hub - Needs your attention card**. Up to three rows pulled by urgency.
- **Hub - Need attention pipeline stat**. The count of unique files with actionable reminders.
- **Reminders page (work queue)**. The full list.
- **File detail Reminders tab**. The reminders attached to one file.
- **File row badges**. A staleness pill shows "Awaiting N days" on the file lists.

Automated client chase

When the work queue would otherwise spawn a manual task, the automated client chase tries to do the chase itself. The daily cron at 08:30 UK Mon-Sat groups due items per contact and sends one digest per contact. The chase tone branches on what the step is sitting on: "Yes this is done" for items the client has to do, "Yes this is already done" for items sitting with the solicitor or lender. If the chase cannot run (the contact has unsubscribed, has no email, has hit the max-chase cap, has been silent for over 14 days, has had emails paused, or has no portal token), the engine creates an agent fallback task instead with a labelled fallback-kind chip.

Notifications outside the reminder engine

- **In-app notifications bell.** The internal staff dashboard has a bell that surfaces unread notifications. The confirmed type today is `portal_milestone_confirm` (a client confirmed a step from the portal). Other notification types fire too but the complete enum was not fully traced.
- **Push notifications.** The portal supports web push via VAPID. The agent app exposes a push subscription mechanism but the live triggers were not exhaustively traced.
- **SendGrid event webhook.** Delivery, deferral, and bounce events come back from SendGrid and are written against the originating queue rows.

The cron index

Twenty-four crons are scheduled in `vercel.json`. They fall into three groups: lifecycle (move sales forwards), maintenance (keep the data clean), and internal insight (feed the founder dashboards).

Cron	Schedule	What it does	Audience
/api/reminders/run	04:00 Mon-Sat	Evaluates every active file against every reminder rule. Creates and updates ReminderLog rows; spawns ChaseTask rows for newly-due logs.	Agents
/api/cron/morning-digest	08:00 Mon-Fri	Daily summary email to the team. INTERNAL	Internal staff and directors
/api/cron/agent-weekly-brief	07:00 Mon	Builds and sends the Monday weekly brief.	Agents
/api/cron/client-weekly-update	09:00 Sat	Weekly reassurance update to vendors and purchasers when their file has been quiet.	Buyers, sellers
/api/cron/check-domains	02:00 daily	Validates SendGrid DNS auth on each agency sending domain. Emails the owner if it breaks.	Agents (alert only)
/api/cron/detect-problems	03:00 daily	Runs the problem-detection scan; writes ProblemFlag rows for stale/anomalous files.	Agents (read via UI)
/api/cron/retention-sweep	09:00 daily	Drives the activation funnel emails (day 1, day 3, day 21, day 30, day 60).	Agents
/api/cron/backfill-mode-profile	04:00 daily	Backfills agency mode profile data. INTERNAL	System only
/api/cron/rollup-metrics	02:00 daily	Yesterday's daily metric row; rolls 13 weeks of WeeklyCohort. INTERNAL	System only
/api/cron/signals	03:00 daily	Anomaly detection across the pipeline. INTERNAL	Command Centre
/api/cron/daily-brief	06:00 daily	AI-generated signal digest for the founder. INTERNAL	Founder
/api/cron/weekly-review	07:00 Mon	AI-generated 7-day signal and experiment review. INTERNAL	Founder
/api/cron/content-topics	04:30 daily	Suggests LinkedIn post topics from recent activity. INTERNAL	Founder
/api/cron/content-batch	06:30 daily	Bundles approved draft posts into an email to the inbox. INTERNAL	Founder
/api/cron/data-retention	01:00 Sun	Anonymises non-superadmin users inactive 3+ years with no active sales.	Compliance
/api/cron/file-time-close	03:00 daily	Closes stale file-time-tracking sessions.	System only
/api/cron/drain-outbound-email	Hourly Mon-Sat	Drains the outbound email queue (everything except step-confirmation digests).	All recipients
/api/cron/send-milestone-digests	Every 3 minutes	High-frequency drain of step-confirmation emails. Groups by recipient and digests when N is at least 2.	Buyers, sellers
/api/cron/drain-withdrawal-notifications	Hourly Mon-Sat	Drains the chain withdrawal notification queue; fires wait-nudge after 14 days of silence.	Agents
/api/cron/medians-ready-check	09:00 daily	One-shot. When 50+ files have completed at least one step, emails superadmins a median-duration comparison. INTERNAL	Founder
/api/cron/client-chase	08:30 Mon-Sat	The automated client chase engine entry point. Enqueues chase digest per contact, runs the escalation pass, and creates agent fallback tasks when chases cannot fire.	Buyers, sellers, agents (fallback)
/api/cron/accrue-invoices	03:00 daily	Per-agency reconciliation of the current month's building invoice against billed exchanges. Applies any unapplied credit notes.	Directors (read), system

/api/cron/check-failed-payments	04:00 daily	Flips agency new-file creation to blocked once a failed payment is 7 days overdue.	Directors, system
/api/cron/issue-invoices	05:00 on the 1st	Flips prior-month building invoices to issued; pushes to Stripe for auto-collection.	Directors, system

2.3 The sale lifecycle and its steps

A sale moves through one of five states and is shaped by two orthogonal axes: **service type** (self managed or outsourced to Sales Progressor) and **tenure plus purchase type** (which determines which steps auto-skip).

The lifecycle states

State	What it means	What the system does
draft	Created but not yet submitted to the live process	Hidden from internal-staff search by default.
active	Live and progressing	Picked up by the reminder engine, the chase cron, the hub stats, the weekly briefs.
on_hold	Paused (e.g. renegotiating, awaiting a client decision)	A TransactionHoldPeriod row is opened. Time-based signals freeze. Reminders stop. Audit fields record who started and ended the hold and an optional planned-end date so expired holds surface again.
completed	Both exchange and completion steps confirmed on both sides	Terminal. Triggers the completion survey.
withdrawn	Fell through	Terminal. Optional <code>fallThroughReason</code> stored. The chain withdrawal cascade fires if the file was linked.

The service types

Service type	Fee at exchange	Who confirms steps
Self managed	£59 per file	The agent themselves
Outsourced to Sales Progressor	£250 (up to £349,999), £300 (£350,000 to £499,999), £350 (£500,000 and above)	The internal sales-progressor team

Auto-skip rules

Three rules determine which steps are auto-marked "not required" at file creation. They re-evaluate when the agent edits the sale details mid-flight.

- **Tenure = freehold:** VM8, VM9, PM12 auto-skip (no management pack on freeholds).
- **Purchase type = cash buyer:** PM5, PM6, PM11 auto-skip (no mortgage application, lender valuation, or mortgage offer).
- **Purchase type = cash from proceeds:** PM5, PM6, PM11 auto-skip, plus PM24 (the deposit transfer step is skipped because the deposit comes out of concurrent sale equity).

The full step list - Vendor side

Code	Name	Blocks exchange	Prerequisite
VM1	Seller has instructed their solicitor	yes	-
VM2	Seller has received the memorandum of sale	yes	-
VM3	Seller has received the welcome pack from their solicitor	yes	VM1
VM4	Seller has completed ID and AML checks with their solicitor	yes	VM3
VM5	Seller has received the property information forms from their solicitor	yes	VM4
VM6	Seller has returned completed property information forms to their solicitor	yes	VM5
VM7	Seller's solicitor has issued the draft contract pack	yes	VM6
VM8	Seller's solicitor has requested the management pack	yes	-
VM9	Seller's solicitor has received the management pack	yes	VM8
VM10	Seller's solicitor has received initial enquiries	yes	VM7
VM11	Seller has provided initial replies to their solicitor	yes	VM10
VM12	Seller's solicitor has issued initial responses to the buyer's solicitor	yes	VM11
VM13	Seller's solicitor has received additional enquiries	yes	VM10
VM14	Seller has provided additional replies to their solicitor	yes	VM13
VM15	Seller's solicitor has issued additional responses to the buyer's solicitor	yes	VM14
VM16	Seller's solicitor has issued contract documents to the seller	yes	VM7
VM17	Seller's solicitor has received signed contract documents back from the seller	yes	VM16
VM18	Seller's solicitor has confirmed readiness to exchange (gate)	no	-
VM19	Seller has received confirmation that contracts have exchanged	no	VM18
VM20	Seller has received confirmation that the sale has completed	no	VM19

The full step list - Purchaser side

Code	Name	Blocks exchange	Prerequisite
PM1	Buyer has instructed their solicitor	yes	-
PM2	Buyer has received the memorandum of sale	yes	-
PM3	Buyer has completed ID and AML checks with their solicitor	yes	PM1
PM4	Buyer has paid money on account to their solicitor	yes	PM1
PM5	Buyer has submitted their mortgage application	yes	-
PM6	Lender valuation has been booked	yes	PM5
PM7	Buyer's solicitor has received the draft contract pack	yes	PM4
PM8	Buyer's solicitor has ordered searches	yes	PM7
PM9	Buyer has booked a Level 2 or Level 3 survey	yes	-
PM10	Buyer has received the survey report	yes	PM9
PM11	Buyer's solicitor has received the mortgage offer	yes	PM6
PM12	Buyer's solicitor has received the management pack from the seller's solicitor	yes	VM9
PM13	Buyer's solicitor has received the search results	yes	PM8
PM14	Buyer's solicitor has raised initial enquiries to the seller's solicitor	yes	PM7
PM15	Buyer's solicitor has received initial replies from the seller's solicitor	yes	PM14
PM16	Buyer's solicitor has reviewed the initial replies	yes	PM15
PM17	Buyer's solicitor has raised additional enquiries	yes	PM14
PM18	Buyer's solicitor has received additional replies	yes	PM17
PM19	Buyer's solicitor has reviewed the additional replies	yes	PM18
PM20	Buyer's solicitor has confirmed all enquiries are now satisfied	yes	PM19
PM21	Buyer has received the final report from their solicitor	yes	PM20
PM22	Buyer's solicitor has issued contract documents to the buyer	yes	PM21
PM23	Buyer's solicitor has received the signed contract documents back from the buyer	yes	PM22
PM24	Buyer has transferred the deposit	yes	PM23
PM25	Buyer's solicitor has confirmed readiness to exchange (gate)	no	-
PM26	Buyer has received confirmation that contracts have exchanged	no	PM25
PM27	Buyer has received confirmation that the sale has completed	no	PM26

What fires when a step is confirmed

Every step confirmation runs an atomic cascade:

1. The step's direct dependents are unlocked.
2. Any reminder logs targeting this step are auto-completed.
3. The exchange gate (VM18 on the vendor side, PM25 on the purchaser side) is re-evaluated: if every blocking step on that side is now complete or not required, the gate unlocks.
4. Any out-of-order completion flags downstream resolve themselves when their prerequisite chain catches up.

5. The file's `lastActivityAt` is touched.
6. An event row is written to the audit ledger.
7. An email is enqueued to the other contacts on the file (single confirmation email, or a digest if more pile up inside three minutes).

The bilateral pairs

Four steps are bilateral: confirming one half of a pair confirms the other inside the same database transaction.

- **VM19 (exchange, vendor side) ↔ PM26 (exchange, purchaser side).**
- **VM20 (completion, vendor side) ↔ PM27 (completion, purchaser side).**

Exchange (the billing trigger)

VM19 or PM26 confirming sets `exchangedAt` on the file. If `freeOnExchange` is true (trial file), the engine stops there. Otherwise the file's purchase price is snapshotted into `priceAtExchange` and `billedAtExchange` is stamped. The snapshot is what the running total and the invoice line use from this point on. Subsequent price edits on the file do not change the fee.

Completion

VM20 or PM27 has no billing side effect (billing was already stamped at exchange). It fires a chain completion notification to claimed chain mates and, if every claimed link in the chain has now completed, enqueues a whole-chain celebration email (one per agent, guarded against duplicates).

Withdrawal and reversal

Setting the file's status to `withdrawn` records the optional fall-through reason, writes an internal note on the activity feed, and fires the chain withdrawal cascade (Lost buyer, Lost purchase, Asked to wait emails) if the file is chain-linked. Note: this does *not* reverse billing. If the file had already exchanged, the fee stays on the invoice.

If the agent instead *undoes* VM19 or PM26, the billing reversal helper fires. If the invoice is still in its "building" state for the current month, the file simply drops off it. If the invoice has already been issued, a credit note is written and applied to the next month's invoice.

2.4 The client portal

Full detail of what buyers and sellers see appears in sections 1.2 and 1.3. This section covers the portal as a system.

Authentication

Token in URL. Each `Contact` row carries a unique `portalToken`. Tokens persist indefinitely; there is no expiry or scheduled rotation in code. Tokens are generated by the agent from the file detail page via the `generatePortalTokenAction` server action; the token is then included in the portal invite email. There is no self-serve "re-request a portal link" mechanism for a contact; the agent has to regenerate.

Per-contact and per-token rate limiting

Portal-invite emails and explain-an-email calls are rate-limited per token to prevent abuse. The explain-an-email feature caps at three calls per contact per hour.

Document uploads

The Home page exposes an upload widget for the survey report. Submissions hit `/api/portal/documents?token=...`. Validation: PDF/JPG/PNG/DOCX, 10MB ceiling, up to three files per upload. Storage in Supabase at `{transactionId}/{timestamp}-{random}.{ext}`. A `TransactionDocument` row is created and a `file_uploaded` event is recorded. The portal itself does not let the user download what they uploaded; that surface is agent-only.

PWA and push

The portal has a Web App Manifest at `/api/portal/manifest/[token]` and supports the install prompt. Web push opt-in is supported via VAPID. The portal sets `viewport: { maximumScale: 1, userScalable: false }` and respects `env(safe-area-inset-bottom)` for notch clearance.

Unsubscribe

Every chase email carries an HMAC-signed unsubscribe link. Clicking sets the contact's `unsubscribedAt` timestamp. The portal stays accessible. The respond page renders an "opt-out" banner but still lets the contact confirm steps, set dates, or leave notes manually. There is no re-subscribe UI; the agent has to flip the flag.

2.5 Chain handling, invites, and claiming

Chain handling is one of the harder problems the app solves and one of the larger code surfaces.

The data model

A **PropertyChain** groups multiple **ChainLink** rows, one per position in the chain. Each link is either a *stub* (just an address and counter-party agent details, no real file behind it) or *claimed* (linked to a real `PropertyTransaction`). A file knows which chain link is its active one via `chainLinkId`.

Inviting the other agent

The originating agent adds a stub for the upward or downward neighbour and presses "invite". The system generates a token (7-day expiry on first send, 14 days on resend) and sends a **chain invite** email to the counter-party agent with two CTAs: Claim and Decline. The chain invite email lists the originator's agency, both property addresses, and the chain position.

Declining

Clicking Decline lands on a public page and writes back to the link without authentication. A decline notification fires to the originating agent.

Claiming

Clicking Claim lands on `/claim`. If the counter-party already has an account they sign in; if not they sign up via `/claim/signup` (which posts to `/api/register` with a `claimSignup` flag so the generic activation email is suppressed). After authentication the form posts to `/api/claim` with one of two actions:

- **create** - a new sale is created on the counter-party's agency from the stub data.
- **link** - an existing sale on the counter-party's agency is linked to the chain.

Either action verifies the user's email matches the stub email, asserts the new agency can create files (payment block check), stamps `firstSubmissionAt` if this is the agency's first sale, sets the link's `transactionId`, `claimedByUserId`, `claimedAt`, and flips the invite status to CLAIMED. The **claim welcome** email then fires - a property-aware welcome distinct from the generic activation email.

Reconciling history at claim

If the counter-party indicates "already in progress", they can retroactively mark steps as completed on the new file at the point of claim (e.g. "exchanged 3 days ago"). The reconciliation runs fire-and-forget after the main claim commits.

Step notifications across the chain

When VM19, PM26, VM20, or PM27 fires on any chain-linked file, the engine looks up every other CLAIMED link on the same chain (excluding the originating link) and enqueues a chain exchange or completion notification to each one. Recipients with suppressed addresses or per-user chain-email opt-outs are skipped. The deduplication key is (sourceId, recipient, emailType) so a duplicate notification cannot fire for the same chain event.

Withdrawal cascade

When a chain-linked file is withdrawn, `cascadeChainWithdrawal` queues:

- **Lost buyer** to the nearest upward claimed mate.
- **Lost purchase** to the nearest downward claimed mate.
- **Asked to wait** if the chain is being reformed.

Recipients can respond with REMARKETING (will wait), WITHDRAW, or implicitly WAITING. If 14 days pass with no response, a single **wait nudge** fires. There is no further chasing after the nudge.

Whole-chain celebration

When every claimed link in the chain has reached completion, one celebration email goes to each claimed agent. Guarded by `chain.celebrationSentAt` so it cannot fire twice.

2.6 Billing and charging

Sales Progressor is **not VAT registered today**. Every price quoted in the app is the actual price. Charging fires on **exchange**, not on completion.

The fee schedule

Service	Fee	Notes
Self managed (in-house file)	£59 per file	The agent confirms steps themselves.
Outsourced - up to £349,999	£250 per file	Banded on the purchase price snapshot at exchange.
Outsourced - £350,000 to £499,999	£300 per file	
Outsourced - £500,000 and above	£350 per file	

Source of truth: `Lib/billing/fee.ts`. The VAT-split helper in the same file is dormant today; when Sales Progressor registers for VAT the split is automatically applied.

The trial

Every new agency gets a **14-day trial**. The clock starts on the agency's first sale (`firstSubmissionAt` is stamped at that moment). Any file that exchanges inside that 14-day window has its `freeOnExchange` flag set true at creation time and is exchanged without billing. The file still appears on the running total page, surfaced as "Free - trial" with the fee that would have applied so the director can see what the trial gave away.

The director's running total

The billing page at `/agent/account/billing` shows:

- This month's metrics strip: total charges so far, exchange count, in-house versus outsourced split, lifetime trial savings, lifetime billed amount.
- The current month's building invoice with line items (billed exchanges, trial exchanges marked free, pending credits). A "preview PDF" button renders the building invoice via the server-side PDF helper.
- The history of past issued invoices with status (issued, paid, failed) and totals.
- The payment-method panel, with the right action surfaced based on state: pending (terms not seeded), add card (acknowledged but no Stripe customer), card on file (Stripe customer attached).
- Trial state and acknowledgement status.

The acknowledgement gate

Before the director can capture a payment method they must read and acknowledge the active terms version. The `TermsVersion` table holds the canonical commercial-terms document. The latest row with `effectiveFrom ≤ now` is the active version. The director's page renders a four-state machine for the payment-method panel:

1. **stripe_not_configured** - the env vars are missing. Should not happen in production.
2. **pending** - no terms version is seeded.

3. **disclosure** - terms exist, the director has not yet clicked acknowledge.

4. **card_form** - acknowledged, ready for Stripe card capture.

The acknowledgement audit row denormalises the user's name and email at the point of acknowledgement for GDPR robustness. When the founder publishes a new terms version with a later `effectiveFrom`, a fresh acknowledgement is required.

How invoices accrue and issue

Stage	What happens	Cron
Exchange	The file's price is snapshotted into <code>priceAtExchange</code> . <code>billedAtExchange</code> is stamped. Bilateral pair re-entry is guarded.	None - fires synchronously inside the step confirmation
Daily accrual	For each agency with billed exchanges in the current Europe/London month, the open "building" invoice is reconciled against the billed-exchange rows. Orphan lines (from reversals) are removed; new lines are added; unapplied credit notes are applied as <code>credit_applied</code> lines (negative).	<code>/api/cron/accrue-invoices</code> , 03:00 daily
Monthly issuance	On the 1st of each month, every prior-month building invoice flips to "issued". A Stripe invoice is created with <code>collection_method=charge_automatically</code> and the Stripe invoice ID is persisted. Zero-net invoices are marked issued without a Stripe call.	<code>/api/cron/issue-invoices</code> , 05:00 on the 1st
Stripe payment webhook	Successful payment marks the invoice paid and clears any agency-level failure flags. Failure marks the invoice failed and stamps <code>paymentFailedAt</code> on the agency once per month.	Webhook
Failed-payment guardrail	If the failed-payment timestamp is more than 7 days old, the agency's <code>newFileCreationBlockedAt</code> is set. New sales and chain claim "create" actions are then blocked until payment succeeds.	<code>/api/cron/check-failed-payments</code> , 04:00 daily

Credit notes and reversals

If VM19 or PM26 is undone *after* the file's invoice has been issued, a `CreditNote` is written for the full gross fee, marked as unapplied. The next accrual run picks it up and writes a `credit_applied` line on the agency's current building invoice. If the file is undone *before* issuance (still in the same calendar month), the file simply drops off the building invoice and no credit note is needed.

Mid-flight service-type switching

An admin can flip a file from self managed to outsourced or back via the pill in the file's header. The action also clears the assigned progressor user when switching to self managed, sets `progressedBy`, and writes an internal note on the activity feed. Reminders, hub bucketing, and billing all read `serviceType` live, so the new behaviour applies from the moment of the switch.

Lifetime metrics

Lifetime totals shown on the billing page read from issued and paid invoice lines (history), not from a live recompute. Trial-savings totals are read from `PropertyTransaction` rows with `freeOnExchange = true` and `exchangedAt` set, computing what would have been charged via `computeFee()`.

2.7 Accounts and access

Sign-up

An agency creates itself through one of three paths:

- **Self-serve password** at `/signup`. The form posts to `/api/register`, which validates the inputs and calls `createDirectorWithAgency`, atomically creating the Agency row and the first User. Password is bcrypted before storage. The generic activation email is sent on success unless this is a claim signup.
- **Google OAuth** via NextAuth. The Prisma adapter creates the user on first sign-in. The user is then routed to `/signup/complete` to fill in agency name and role.
- **Azure AD OAuth** via NextAuth. Same flow as Google.

OAuth providers use `allowDangerousEmailAccountLinking: true` because both Google and Microsoft verify email ownership before issuing the token. The Account table links the OAuth identity to the User row.

Inviting team members

Direction	Trigger	Email	Expiry
Director invites negotiator	Director adds a negotiator on the team page	Negotiator invitation email with accept link	7 days; resend extends
Negotiator invites director	Only possible if the agency has no director	Director invitation email with accept link	7 days; resend extends

The invitee lands at `/invite-negotiator/[token]` or `/invite/[token]` and can accept via password or OAuth. Accept atomically creates the user, marks the invitation as accepted, and inherits the inviter's `firmName`. Both invitation models support cancel and resend.

Authentication

- NextAuth with three providers: Credentials, Google, Azure AD.
- JWT session strategy. The jwt callback populates user id, role, agencyId, firmName, and a `needsSignupCompletion` flag.
- Credentials authorize is rate-limited per IP via `checkAuthLimit`.
- Login success and failure events are written to the Event audit ledger with IP metadata.

Password reset

POST to `/api/auth/forgot-password` generates a 1-hour-expiry VerificationToken keyed on (email, token) and sends the reset email. The endpoint always returns 200 to avoid email enumeration. The reset link lands at `/reset-password?token=...&email=...` which posts to `/api/auth/reset-password` to apply the new bcrypted password and delete the token.

Multi-firm under one agency

Two soft-scoping fields control visibility:

- **firmName** on User and Agency. Inherited from the director when a negotiator accepts an invitation. Currently only used in the chain invite email subject and a couple of display surfaces; does not drive file visibility.
- **canViewAllFiles** on User. Set by directors on a per-negotiator basis. A negotiator with the flag set sees the full agency pipeline rather than just their own files. The flag is read by every list query and write guard.

Visibility by role

The canonical helper is `lib/security/access-scope.ts`. It returns one of three shapes:

- **all** - admin and superadmin see every file across every agency.
- **assigned** - sales progressor sees only files where `assignedUserId` matches.
- **agency** - director and negotiator see files in their agency (negotiators further scoped to their own unless `canViewAllFiles` is set).

Audit ledger

Every meaningful action is written to the `Event` table via `recordEvent`. The helper is fire-and-forget so a logging failure cannot break the user action. EventType categories include auth events (login, password reset, invite, accept), agency events (created, mode changed, archived), transaction events (created, archived, status changed), step events (confirmed, marked not required, reversed, exchange gate unlocked, contracts exchanged, sale completed), comms events (chase sent, chase message generated, email parse attempted), file events (uploaded, deleted), feedback (submitted), and admin events (admin login, admin action performed).

Data retention

The data-retention cron runs every Sunday at 01:00. It anonymises non-superadmin users with `updatedAt` older than 3 years who have no active sessions and no active or on-hold sales. Email is replaced with `anon-{id}@deleted.invalid`, name is replaced with "Anonymised User", and personal fields (phone, password hash, TOTP secret) are nulled. A `?dryRun=true` query parameter returns eligible users without writing.

2.8 Data, integrations, and app extras

Integrations

Service	What it does in the app	Status
Supabase (Postgres + Storage)	Primary database; portal file uploads	Live
SendGrid	All outbound email; bounce/deferral/delivery webhook	Live
Anthropic Claude	Chase message generation, content drafting, explain-an-email, founder insights	Live
Stripe	Payment method capture and invoice collection for agency billing	Live
Vercel cron	All 24 scheduled jobs	Live
Land Registry SPARQL	Property price history shown in the property-intel panel	Live
EPC Register	Energy performance data on the property panel	Live
Replicate (FLUX)	AI image generation for marketing posts INTERNAL	Pending API token
PostHog EU	Product analytics	Pending DPA + key
Upstash Redis	Rate limiting	Pending account; feature-flagged off

Multi-tenancy

Customer-agency data is scoped by `agencyId`. Internal staff (admin, sales progressor, superadmin) have `agencyId = null` and are scoped through the access-scope helper instead. Every multi-tenant read or write goes through `scopeTransactionWhere`, `scopeOwnershipWhere`, or one of the relation-specific scope helpers.

Push notifications and PWA

The portal has full PWA manifest support and Web Push opt-in via VAPID. The agent app exposes a push subscription mechanism but the full set of in-product push triggers was not exhaustively traced and is listed in the "needs verifying" appendix.

Appendix A - Internal roles

For completeness: what Sales Progressor's own staff get. Not for the customer-facing inventory.

The three internal roles

Role	Surface	Brief
sales_progressor	The agent app, scoped to assigned files	The progression team. Confirms steps on outsourced files on behalf of agents.
admin	The agent app, scoped to all files	Founder / senior internal. Sees every agency's files.
superadmin	Command Centre at /command/*	Founder. The only role with access to the app's own internal analytics, the audit trail, and the experiment manager.

The Command Centre INTERNAL

Page	Purpose
/command/overview	Platform-health snapshot. KPI deltas, "what changed" widget, mode toggle, agency filter, severity badges, recent deployments.
/command/health	Job status table. Recent job runs. Deployment history.
/command/activity	Raw transactional event log.
/command/audit	Audit trail of data changes per resource.
/command/automations	Client-chase automation overview and per-step engagement stats.
/command/content	Editorial content hub. Draft and published templates and guidance.
/command/content/topics	Topic taxonomy and per-step copy assignments.
/command/content/voice	Brand voice guidelines.
/command/content/engagement/[id]	Per-draft engagement analytics.
/command/experiments	A/B test manager.
/command/retention	Churn analysis and retention levers per agency.
/command/growth	Activation funnel, self-serve onboarding, expansion revenue.
/command/friction	Feature adoption, error rates, usage gaps.
/command/insights	Data-warehouse insights and custom dashboards.
/command/outbound	Outbound marketing hub. Campaigns, messaging variants.
/command/revenue	MRR, ARR, churn, LTV, CAC.
/command/setup-2fa, /command/auth/step-up	TOTP setup and step-up verification before protected routes.

Internal-only emails INTERNAL

Email	Trigger	Recipient	Purpose
Morning digest	08:00 Mon-Fri	Sales progressors, admin, directors per agency	Daily file summary, exchange forecast, momentum.
Daily founder brief	06:00 daily	Founder	AI-generated signal digest.
Weekly founder review	07:00 Monday	Founder	AI-generated 7-day signal and experiment review.
Medians ready	09:00 daily (one-shot)	Superadmins	50-file threshold crossed; per-step median-duration comparison.
Content batch	06:30 daily	Founder inbox	Bundles approved draft posts.

Service-type switching INTERNAL

Admin only. The pill on the file header opens a confirm-and-switch modal that posts to `switchServiceTypeAction`. The action only fires for active files; completed, withdrawn, draft, and on-hold files are deliberately not switchable through the UI. Activity-feed audit row is written.

Appendix B - Needs verifying

Items the sweep surfaced but could not fully confirm from the code. The next pass should resolve these.

1. **Notification bell - full enum of types.** The audit confirmed `portal_milestone_confirm` and "portal chase note" as notification types. Other notification kinds are referenced in code but the complete enum was not exhaustively traced. Recommend a focused sweep of every call to `notifyPortal*` and `notify*` in `lib/services` to lock the list.
2. **Push notification trigger inventory.** The agent app sets up a push-subscription mechanism but the complete set of in-product push events was not traced. Likely small (probably a handful of escalation and portal-confirm events) but should be confirmed.
3. **Backfill mode profile cron.** The cron exists at `/api/cron/backfill-mode-profile` and runs daily at 04:00. What it backfills and against what data was not fully traced.
4. **Daily brief and weekly review output shape.** The internal-founder briefs are confirmed to fire but whether they email, render to the Command Centre dashboard, or both was not fully traced.
5. **Mid-flight service-type switch and the fee at exchange.** The fee helper takes the live `serviceType` from the file at billing time. If an admin flips the service type *after* the file has exchanged, the accrual recomputes the fee against the new service type. There is no `serviceTypeAtExchange` snapshot equivalent to the `priceAtExchange` snapshot. Confirm whether this is intentional or a gap to close.
6. **Portal token rotation policy.** Tokens persist indefinitely with no automatic rotation. If a token leaks the agent must regenerate manually. Confirm whether a rotation cron, a user-side "request a new link" button, or a TTL is wanted.
7. **Two-way portal messaging.** The portal supports "leave a note" on the respond page (one-way to the agent) and the Updates tab shows messages from the agent (read only). Full two-way messaging is not implemented. Confirm whether this is the intent.
8. **Bilateral chain notifications for steps other than exchange and completion.** The schema reserves capacity for bilateral coordination on other step pairs. The current code only fires on VM19/PM26 and VM20/PM27. Confirm whether more bilateral pairs are planned.
9. **GDPR hard delete.** The data-retention cron anonymises after 3 years. Comments in code reference a `/api/gdpr/delete` hard-delete endpoint but it was not located in the current tree. Confirm whether it exists and what it does.
10. **Email verification on signup.** The `User.emailVerified` field exists but is populated only by NextAuth OAuth. Password sign-ups do not currently require email verification. Confirm whether that should change.
11. **Director-only billing UI - the polish vs production paths.** Two billing UI routes were found: `/agent/account/billing` and `/agent/polish/billing-hub-v2`. Confirm which is live in production and remove the other from the doc if appropriate.
12. **The drain crons skip Sunday on most queues.** Drain-outbound-email and drain-withdrawal-notifications both run hourly Mon-Sat. Confirm this is intentional (UK weekend operations) versus a configuration drift.
13. **Outsourced-fee bands and "legacy" agencies.** The agent-admin page mentions an Agency fee override for "legacy" outsourced files. The override mechanism was referenced but not fully audited. Confirm what an override does to the standard banded fee.

14. **The polish and preview routes.** A handful of routes under `/agent/polish/*`, `/agent/audit/*`, `/agent/system-preview/*`, `/agent/anim-preview`, `/agent/hub-preview`, `/agent/analytics-preview` exist as polish or staging surfaces. They are not customer-facing and have not been listed individually in Part 1. Confirm they should stay out of the customer-facing inventory.

End of document. Reflects commit `3dea586`, dated 3 June 2026.